

// Day 27 of 30

ARCHITECTURE BREAKDOWN

How Netflix Handles 250M Users

→ Swipe through all 6 slides



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

1/8

The Scale

Staggering numbers

250 million subscribers

190+ countries

17,000+ titles

100M+ hours watched per day

Peak traffic:

- 400 Gbps of video streaming
- Thousands of microservices
- Runs almost entirely on AWS

💡 Netflix = largest microservices deployment in the world



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Microservices — 1000+ Services

Each team owns their services

API Gateway (Zuul/Spring Cloud Gateway)

→ Routes to correct service

Services:

→ User Profile Service

→ Recommendation Engine

→ Video Encoding Service

→ Billing Service

→ Content Delivery

→ A/B Testing Service

Each: own DB, own team, own deploy

💡 Hundreds of deploys per day. No downtime.



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Chaos Engineering

Netflix INTENTIONALLY breaks production

Chaos Monkey:

- Randomly kills servers in production
- Forces teams to build resilient systems

Why?

- 'If your system can't handle random failures, it can't handle real ones'

Results:

- 99.99% uptime
- Systems self-heal automatically

💡 Netflix invented Chaos Engineering



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Content Delivery — CDN

Video from the nearest server

Open Connect CDN:

- Netflix's own CDN
- Servers in ISP data centers
- Video cached at edge
- User streams from nearest server

Result:

- Buffer-free streaming
- Reduced internet traffic by 30%
- Works even with slow connections

💡 Netflix built their OWN CDN. That's commitment.



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Recommendation Engine

How it knows what you want

80% of what you watch =
recommended by the algorithm

Inputs:

- Watch history
- Time of day
- Device type
- What similar users watched
- How long you watched
- What you searched

ML models run in real-time
Personalized for each user

💡 Recommendations save Netflix \$1B/year in retention



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

Key Takeaways for Backend Devs

What we can learn

1. Build for failure (Chaos Engineering)
2. Microservices enable team autonomy
3. Cache at the edge (CDN)
4. Circuit breakers prevent cascading
5. A/B test everything
6. Observability is non-negotiable
7. Automate deployments completely

💡 You don't need Netflix scale to use Netflix patterns



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)

done = true

Follow for more!

Like · Comment · Repost

Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)



Shubham Bhati

[linkedin.com/in/bhatishubham](https://www.linkedin.com/in/bhatishubham)